SQL

→ SQL stands for Structured Query Language.

→ SQL is Relational Database Management System to handel structured data.

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Database

Database is collection of data in a format that can be easily accessed

A software application used to manage our DB is called DBMS [Database management system]

Types of Database

Relational (RDBMS)

- Data stored in tables

Example : MySQL , ORACLE , SQL server Microsoft

Non - relational (NoSQL)

- data not stored in table

Example : mongoDB

What is SQL ?

SQL is a programming Language used to ~~interact~~ ~~whit~~ interact with relational database.

It is used to perform CRUD operatons :

C = Create
R = Read
U = Update
D = Delete

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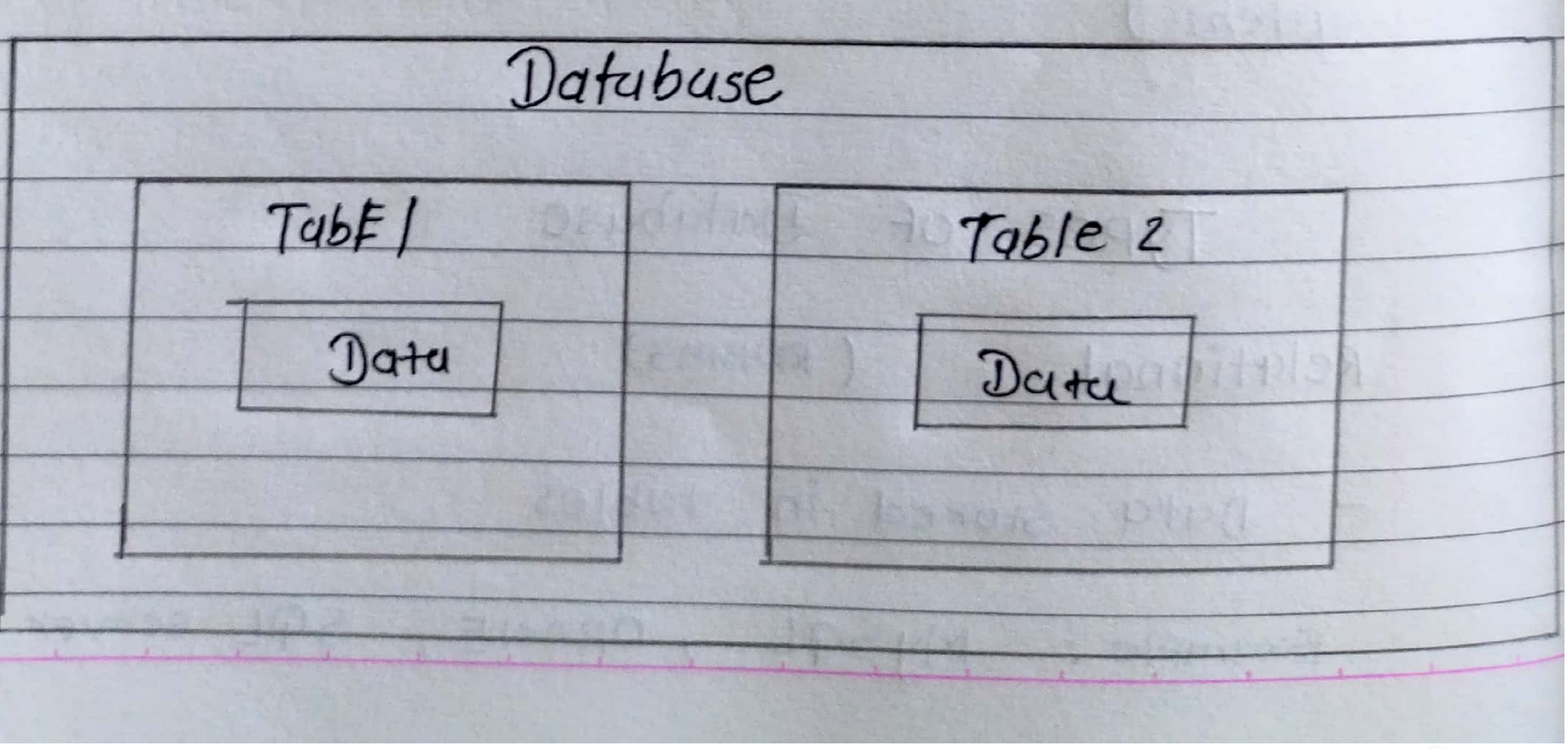
Database Structure



What is a table ?

table is a rows and columns combination.

Creating our First Database

- Our First SQL Query

Creating Database Db-name;
↳ Full Stop

Drop Database Db-name;
↳ Delet Database

Example :

CREAT DATABASE college;
↳ Database-Name

Drop DATABASE college;

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- USE Database query

USE Database_name ;

The use command is used when there are multiple database in the SQl and the user or programmer specifically wants to use a particular database.

Example :

USE college ;

Creating our First table

- Create Table table name (
  Column_name1 datatype constraint,
  Column_name2 datatype constraint,
  );

Example :

```
CREATE TABIE student (
        id INT primary key,
        Sname varchar (50),
);
```


id	sname
.....	.....
.....	.....

- INSERT INTO Quary

The INSERT INTO statement is used to insert new record in a table.

SQL Datatypes

They define the type of values that can be stored in a column.

Datatype

CHAR
VARCHAR
BLOB
INT
TINYINT
BIGINT
BIT
FLOAT
BOOLEAN
DATE
YEAR

Signed & Unsigned

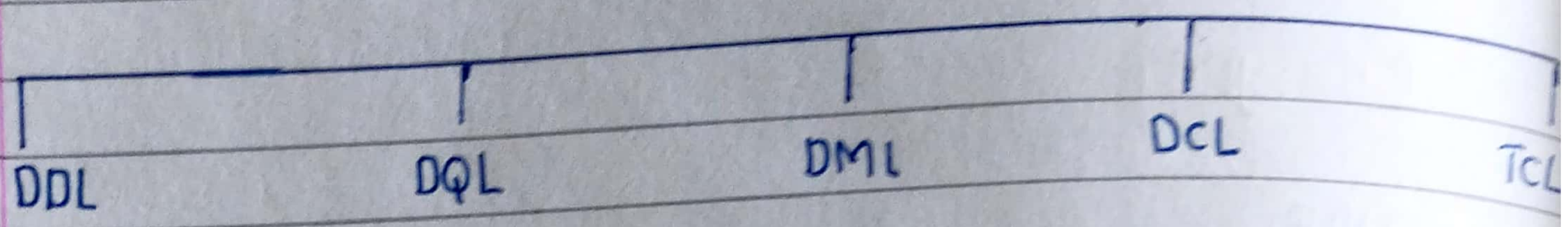
TINYINT UNSIGNED (0 to 255)

TINYINT (-128 to 127)

- unsigned can store only zero and possitive numbers.

- Signed allow zero, positive and negative number.

Types of SQL Commands



DDL : (Data Definition Language) : Create, alter, rename, truncate & drop

DQL (Data Query Language) : select

DML (Data manipulation Language) : Insert, update & delet

DCL (Data control Language) : grant & revok permission to user

TCL (Transaction control Language) : start transaction , commit , rollback E

Database related Query

- CREATE DATABASE db-name ;
- CREATE DATABASE IF NOT EXISTS db-name ;
- SHOW Database ;
- Show Tables ;

Table related command's

- CREATE

```
CREATE TABIE student (
    id INT primary key,
    name varchar (50),
);
```

- Select & view All columns

```
SELECT * FROM table_name;
```

- Insert

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```
INSERT INTO table-name
 ( column1 , column2 )
values
 ( col1-v1 , col2 - v1 ),
 ( col1 - v2 , col2 , v2);
```

- Drop (Delete Table)

```
DROP TABIE table-name;
```

- It's statement is used to drop an existing table in a database.

Example:

```
DROP TABIE student;
```

practice Qs 1

Q.s Create a database for your company named programming.

Step 1 : create a table inside this DB to store student info [id, name and age).

Step 2 : Add following information in the DB:

1, "vijay", 17
2, "SYAM", 18
3, "Ajay", 15

Step 3 : select & view all your table data.

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——————»

```
CREATE DATABASE programming ;
USE programming ;

CREATE TABLE student (
        id INT Primary key,
        name VARCHAR (100),
     age salary INT
);

INSERT INTO student
( id, name, age )
 VALUES
( 1, " vijay ", 17 ),
( 2, "  Syam ", 18 ),
```

(3, " Ajay " , 15) ;

SELECT * FROM Student ;

Keys

primary key :

It is a column in a table that uniquely identifies each row. (a unique id)

There is only 1 primary key & it should be NOT null.

Foreign key

A Foreign key is a column in a table that refers to the primary key

There can be multiple F1K.

FKs can have duplicate & null values.

Example :

table 1 - Student

id (primary key)	citid (foreign key)	city
102	1	pune
101	2	mumbi
103	1	pune

table 2 - city

id (primary key)	city
1	pune
2	mumbai
3	Delhi

Constraints

SQL constraints are used to specify rules for data in a table.

NOT NULL	columns cannot have a null value
UNIQUE	all values in column are different
primary key	makes a column unique & not null but used only for one
foreign key	prevent actions that sh would destroy links between tables
DEFAULT	sets the default value of a column
CHECK	It can limit the values allowed in a column

Syntax :


```
create table Name (
        id INT Primary key,
        Name varchar (50),
        age INT,
        constraint age_check CHECK
          (age >= 18 AND Name="
                RAHUL" )
);

create table new (
        age INT CHECK (age >=18)
);
```

Select in Detail

used to select any data from the database

Basic Syntax

```
select col1, col2 From table-name ;
```

TO select all

```
select * from table-name ;
```

Where clause

To define some conditions

Syntax :

```
SELECT col1, col2 FROM table-name
WHERE Condition ;
```

Example :

```
SELECT * From student WHERE marks > 85;
```

Using operators in WHERE

Arithmetic OP : +, -, *, /, %

Comparison OP : =, <, >, <=, >=, !=

Logical OP : AND, OR, NOT, IN, BETWEEN, All,
like, ANY

Operators

AND to check for both condition to be true

OR to check for one of the condition to be tru

Between select for a given range
(select * from student
where mark Between 80 and 90;)

In matches any value in the list
(same in ("Delhi", "mumbai");)

NOT to negate the given condition

Limite Clause

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sets an upper limit on number of rows to be returned

```
SELECT * From student LiMit 3;
```

Order By clause

(ASC) (DESC)

To short in acending or descending order

```
SELECT * FROM student
order By city Asc;
```

Aggregate Functions

Aggregate function perform a calculation on a set of value, and return a single value.

- Count ()
- MAX ()
- MIN ()
- Sum ()
- AVG ()

Example : 1) SEIECt MAX (marks)
From Student ;

2) SEIECt AVG (marks)
From Student ;

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Group By clause

Groups rows that have the same values into Summery rows.

It collect data from multiple record and groups the result by one or more column.

Example :

```
SELECT city, count (name)
From student
Group By city ;
```

Having clause

Similer to where i.e. ~~application~~ Applices some condition on rows.

used when we want to apply any condition after grouping.

Example :

```
select  count (name), city
From  student
Group By  city
Having  max (marks) >90 ;
```


General order

```
select  columns
From  table_name
where  condition
Group By  Columns
Having  condition
Order By  column (s)  ASC
```

Example :

```
select  city
FROM  student
WHERE  grad = " c "
GROUP BY  city
HAVING  MAX (marks)  >= 90
order By  city  DESC ;
```

* Safe mode : set sql_safe_updates = 0;

1 - on
0 - off

Table related Queries

- update (update existing rows)

```
UPDATE   table_name
set   col1 = val1, val2 = col2
WHERE   condition
```

- Delet (to Delete existing rows)

```
Delete From table_name
where   condition ;
```


Cascading for foreign key

on update cascading

When we create a foreign key using update cascade the referencing rows are updated in the child table when the referenced row is updated in the parent table which has a primary key.

on Delete cascade

When we created a ~~function~~ foreign key using this option , it delete the Referencing rows in the child table when the referenced row in deleted in the parent table which has a pk.

- Alter (to change the schema)

ADD column
Alter Table_name
ADD column column_name datatype;

DROP Column
Alter Table Table_name
DROP column column_name;

Rename Table
Alter Table tuble_name;
Rename To new_table_name;

change column (rename)
Alter table table_name
cHange column old_name new_name;

modify column (modify datatype)
ALter Table table_name
Modify col_name new_datatype;

- Truncate (to delete table's data)

Truncate Table table_name;

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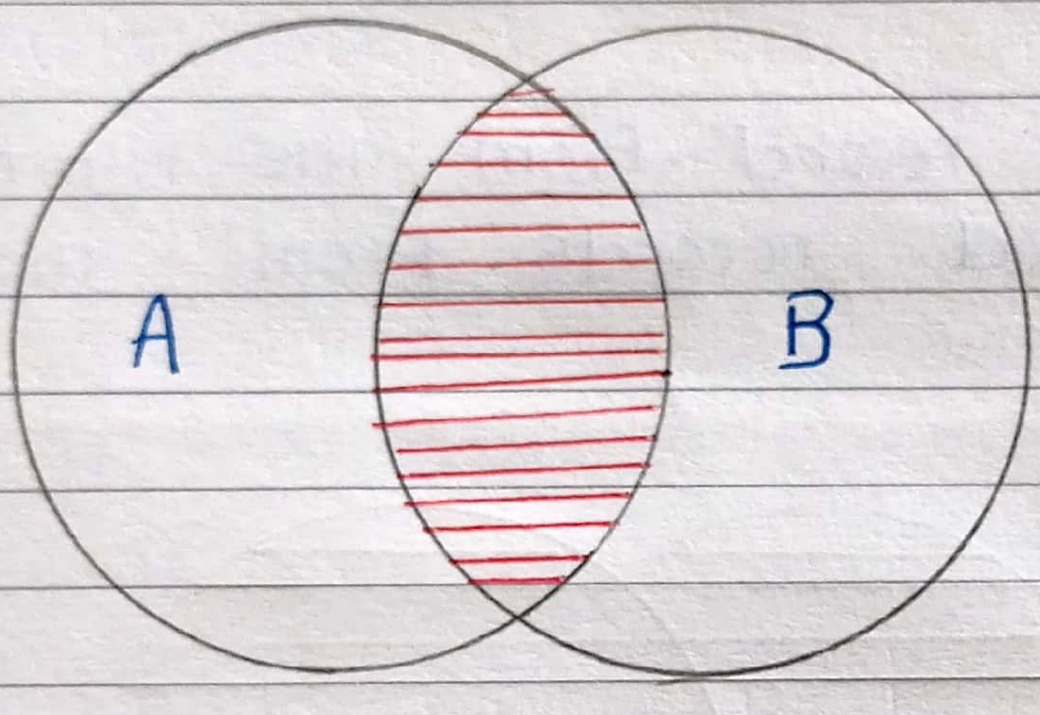
Joins in SQL

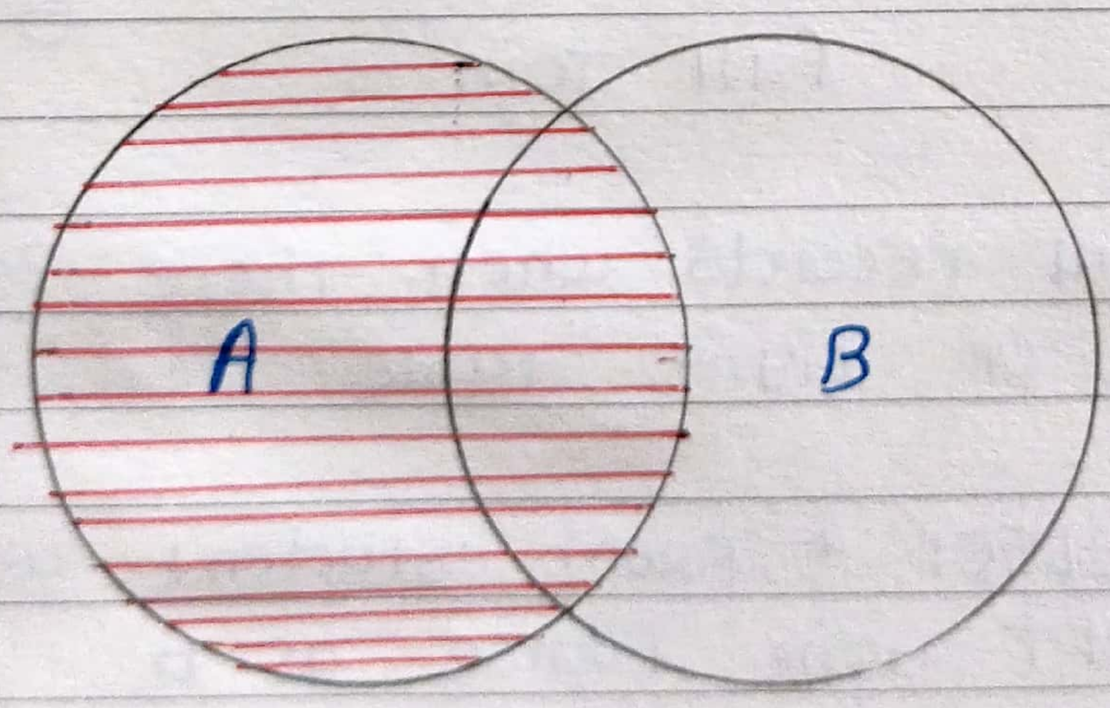
Joins is used to combine row from two or more ~~data~~ tables, based on a releted column between them.

Type's of Joins



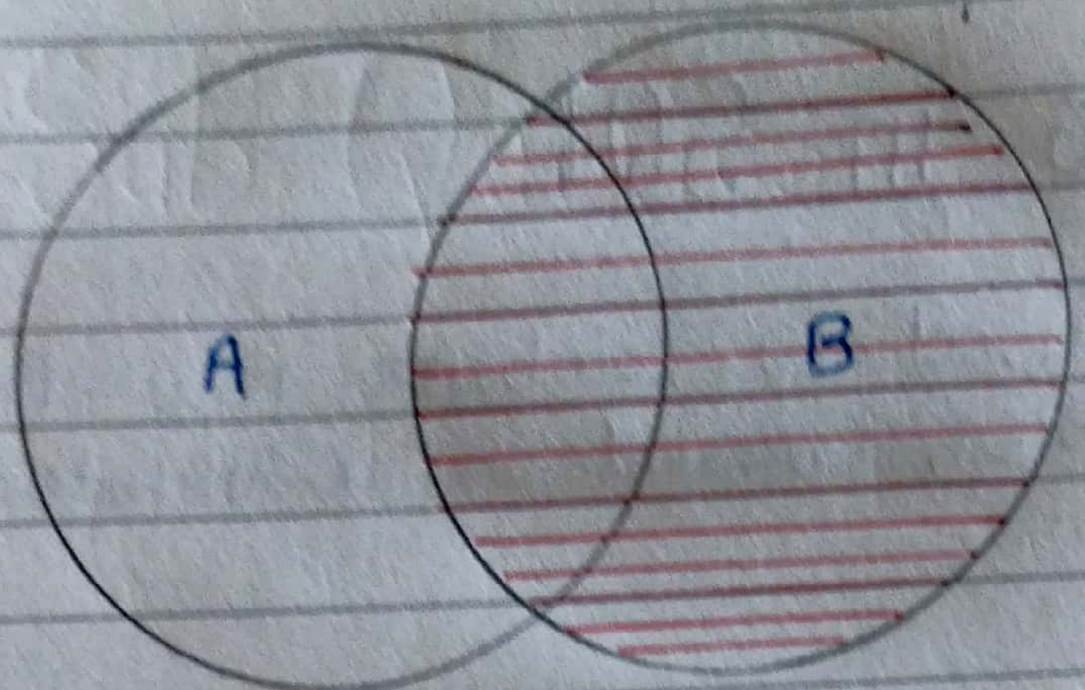
Inner Join

- Return records that have matching value in both tables.



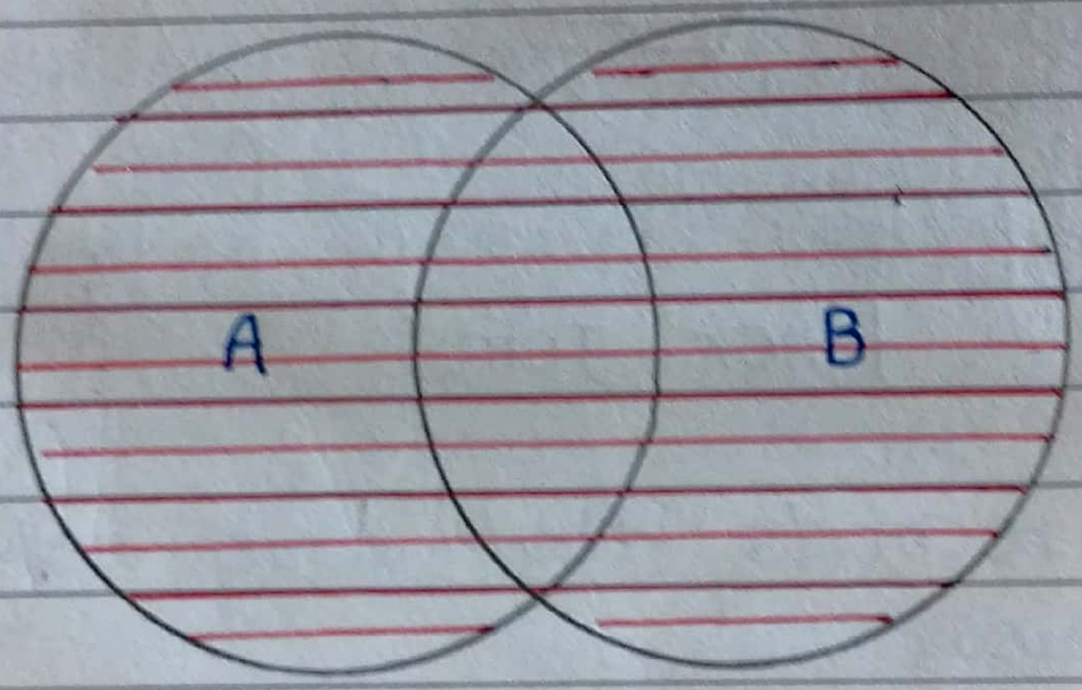
Left Join

- Return all records from the left table, and the matched record from the right table.



Right Join

- Returns all record from the right table, and the mached records From the left table.



Full Join

- Returns all records when there is a match in either left or right table

Syntax :

```
SELECT * FROM student as a
LEFT Join Course as b
on a.id = b.id
union
SELECT * FROM Student as a
RIGHT JOIN course as b
ON a.id = b.id ;
```

LEFT Join
union
Right Join

Self Join

It is a regular Join but the tuble is Joined with itself.

Syntax :

```
SEIECt  column s
From    tuble as u
Join    tuble us b
On  a.col_name  =  b.col_name ;
```

Union

It is a used to combine the result-set of two or more SELECT statements.

Given uniou records.

To use it :

- every select should have sume no. of column
- column must have similar dutu types
- column is every select should be in sume order

Syntax :

```
SELECT  column (s)  FROM  tuble A
union
SELECT  column (s)  FROM  tuble B
```

SQL Sub Queries

A Subquery of Inner query or a Nested query is a query within another SQL que

It involves 2 select statements.

syntax :

```
SEIECT   column (s)
FROM   table-name
WHERE   col-name   operator
(sbquery);
```

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MySQL Views

A view is virtual table based on the resul - self of an SQL statements.

* A view always shows up-to-data data. The database engine recreate the view, every time a user queries it.

Syntax :

```
CREATE   VIEW   Creelit   AS
SEIECT   id,  name   FROM   student-bank;

SEIECT   *  From   creelit
```